Exact-Relation Coding: A Preregistered Test of Whether Discovered Algebraic Structure Yields a Composed Lossless-Compression Advantage

Preregistered empirical study · negative result · experiment state `v0.3-submission`
Repository: https://github.com/Xeloraa/deductive-coding · all numbers traceable to `results/ledger.json`

Author(s): withheld for review — repository: `https://github.com/Xeloraa/deductive-coding` **Version:** experiment state tagged `v0.3-submission`. **Artifact:** all numbers in this paper are generated from `results/ledger.json` (122 experiment records) by `scripts/regen_tables.py`; every inline figure carries a source marker naming the experiment record and field it comes from, verified against the ledger by `scripts/check_paper_numbers.py`.

Abstract

Lossless compressors model a probability for the next symbol and code a residual. We investigate an alternative pre-pass: automatically *discover* an exact algebraic relation the data satisfies, transmit a fully counted description of that relation together with the independent symbols, and let the decoder reconstruct the determined symbols. We formalise the pre-pass as a GF(2) column-basis codec (with an integer-affine variant for tables), define a *composed deduction gain* `G = min_c |c(x)| - min_c |c(D(x))|` over a fixed set of stock compressors `c` and the accounted container `D(x)`, and preregister a per-file success threshold before running any natural-corpus experiment. On data engineered to contain an exact GF(2) linear code the pre-pass removes about half of the size the six stock baselines achieve (1 MiB code: `G` 523,726 bytes, `G_pct` +49.95 %), the gain scales with the number of rows while relation description stays near-constant, and two context-mixing compressors (paq8l, paq8px v216) do not close it. We then test whether comparable exploitable structure transfers to public natural corpora: 8 of 12 whole Silesia members (the largest an 8 GiB machine completes), all 12 Silesia members plus an enwik8 prefix plus six scientific `float32` fields plus a telemetry text file at 256 KiB, and a bounded bit-phase-offset extension of the detector. Under the preregistered protocol, **no natural file produces a composed gain meeting the threshold**; the maximum observed `G_pct` over 122 records is `+0.0009` (a passthrough header artifact), and the offset extension reproduces the phase-0 composed gain to the byte on every file. Functional-dependency and per-record-CRC32 cases reproduce known techniques and are labelled as such. The mechanism is not novel (grammar compression; syndrome coding; recent program-synthesis compression). Our contribution is the accounting-complete codec with an independent verifier, the composed-gain evaluation discipline, and the preregistered empirical finding that blind axis-aligned exact-relation discovery does not yield a composed advantage on the corpora and detectors tested. We do **not** claim natural data contains no algebraic redundancy, nor that the approach can never help.

Keywords: lossless compression, algebraic redundancy, GF(2) linear codes, functional dependencies, preregistration, negative result, reproducibility.

1. Introduction

Redundancy in a byte string is either *statistically predictable* — a good model assigns the next symbol high probability — or *exactly determined* — some relation fixes it given other symbols. General-purpose compressors exploit the first. Whether it is ever worthwhile to explicitly discover and transmit the second, on arbitrary data, after paying for the description, is an old question that is periodically revisited.

It carries a known trap. A "win" measured against the payload alone, or against the transformed container *before* a downstream compressor, routinely evaporates once the container is itself compressed, because the stock compressor already captures the same redundancy given a more convenient byte layout. Any honest test must therefore (i) count every metadata bit inside the container and (ii) compare *after* a strong downstream compressor. We make that comparison the primary quantity and fix it, together with a success threshold, in a git-locked preregistration written before the natural-corpus experiments.

1.1 Research questions

We separate five layers and never let one stand in for another; "we found a relation" (RQ-A) is never reported as a compression result.

- **RQ-A — does exact structure exist?** Does `x` satisfy an exact GF(2)/affine relation at a tried bit width and phase?
- **RQ-B — can we discover and represent it?** Is the relation verified on *every* row and encoded into a well-formed, fully accounted container?
- **RQ-C — does it reduce total representation cost?** Is the container strictly smaller than both a never-worse passthrough and the best stock compressor on the raw bytes, *before* any downstream compressor?
- **RQ-D — does the reduction survive a strong downstream compressor?** Is the composed gain `G > 0`?
- **RQ-E — does it occur naturally at meaningful scale?** Across the preregistered natural corpus, does any non-prior-art file clear the fixed threshold?

RQ-A → RQ-B → RQ-C → RQ-D is a strict per-file funnel; RQ-E is the corpus-level aggregate.

1.2 Contributions

1. A GF(2) column-basis lossless codec with an integer-affine variant, a bit-exact accounting ledger whose finaliser refuses to emit a container whose category bits do not sum to its byte length, and a structural never-worse passthrough fallback (§3).
2. The *composed deduction gain* metric and a preregistered per-file threshold, as an evaluation discipline for pre-pass compression claims (§2, §4).
3. An independent, shared-nothing decoder and end-to-end composed round-trip verification (§5.4).
4. A control battery that makes a negative result *informative*: planted positive, i.i.d./shuffled/near-relation nulls, a corruption sweep, a representation-change false-positive estimate, a metadata-cost check, a composition-order check, and a non-aligned-period scope control (§6).
5. A preregistered empirical finding: on 8 whole Silesia members, 20 further files at 256 KiB, and a bounded bit-phase-offset extension, blind axis-aligned exact-relation discovery yields **no** composed advantage meeting the threshold, robustly to bit phase (§5).

The mechanism itself is not claimed as novel; see §7.

2. Problem definition and the composed metric

Let `x` be the input byte string and `B` a fixed set of lossless stock compressors (§5.2). Let `E` be a lossless deductive encoder with `E^{-1}(E(x)) = x`, and write `D(x) = E(x)` for the *container*. `E` emits the byte-smaller of a relation container (§3) and *passthrough* (`magic · version · kind · len64 · crc32 · x`, all whole-byte fields, so `|passthrough(x)| = |x| + 18`), hence `|D(x)| ≤ |x| + 18` always.

```
raw_best(x)      = min_{c ∈ B, c ran}  |c(x)|
composed_best(x) = min_{c ∈ B, c ran}  |c(D(x))|
G(x)             = raw_best(x) − composed_best(x)         (signed bytes)
G_pct(x)         = G(x) / raw_best(x)
```

`G(x) > 0` is the event the hypothesis needs. A companion *raw gain* `raw_best(x) − |D(x)|` (no downstream compressor) is reported to expose the case where a raw win vanishes under composition. When a compressor fails on one side only (e.g. `xz` runs out of memory on a large container but not on the raw bytes), a *matched-codec gain* over the intersection of compressors that ran on both sides is also recorded; a positive is reportable only if the matched gain also clears the threshold.

3. Method

3.1 GF(2) fixed-width column basis

For a bit width `w ∈ {8, 16, 32, 64, 128, 256}`, the bits of `x` are reshaped to an `⌊8|x|/w⌋ × w` matrix; the `< w` trailing bits are carried verbatim (`leftover`). Gaussian elimination over GF(2) on packed 64-bit rows computes the leftmost column basis: the pivot columns are the unique leftmost independent set, and every free column is expressed as an XOR of pivot columns. The map is checked against the *original* matrix before use — a discovery bug raises an error rather than producing a wrong container. Homogeneous and affine (`[1 | A]`, the all-ones column synthesised at decode and never transmitted) variants are both tried, and the smallest fully accounted container over all `(w, variant)` is kept, else passthrough.

3.2 Integer affine functional dependencies

For byte strings that parse as an `int64` table, candidate relations `z = Σ aᵢ xᵢ + b` are solved on a few rows and verified with exact Python integers on *every* row; coefficients are zig-zag varints, counted. This variant is used only for synthetic functional-dependency controls; the natural-corpus experiments use §3.1 exclusively.

3.3 Bit-phase-offset extension

The detectors of §3.1–3.2 are *axis-aligned* and reshape from bit 0. As the one bounded detector-broadening attempt required by the preregistration's kill criterion, we add a phase sweep: for each width `w`, also reshape starting at every bit phase `p ∈ {0, …, w-1}` (a coarse `p`-subset for `w ∈ {128, 256}`), carrying the skipped `p` bits as a counted `prefix` field. This tests whether a phase-0 negative is a *framing* artifact — a genuine `w`-periodic linear relation beginning mid-byte is invisible to phase-0 reshaping and recovered at the right phase (verified: a planted width-24 code with 8 junk bits prepended is found at `w = 24, p = 8` and missed at phase 0). It is bounded — `Σ w` extra reshapes, no search over nonlinear forms — and does not attempt a learned permutation.

3.4 Container and bit accounting

The relation container is (LSB-first, then byte padding):

```
magic[4] version[8] kind[8]
n_rows[32] n_cols[32] n_payload_pivots[32]
flags[8]   (bit0 affine, bit1 ones-is-pivot, bit2 has-prefix)
original_nbytes[64] leftover_nbits[32]  leftover bits
prefix_nbits[32] + prefix bits           (only if flags bit2)
pivot_mask[n_cols]
per free column, in index order: coefficients[n_payload_pivots (+1 if affine)]
pivot payload: n_rows · n_payload_pivots bits  (row-major, pivot-column order)
crc32(x)[32]
padding to a whole byte
```

Every emitted bit is attributed by `AccountedWriter` to exactly one of `{header, relation, payload, leftover, prefix, crc, framing}`. The finaliser raises unless

```
8·|D(x)|  =  Σ category bits,   framing = (−Σ others) mod 8 ∈ [0, 7].
```

A container with an unaccounted bit cannot be emitted, and a mis-typed category name still counts toward the total (it lands in an overflow bucket), so the invariant cannot be defeated by a typo. `crc32(x)` is written and re-checked on decode; it is not needed for reconstruction, so it makes the deductive container *larger* (conservative for the hypothesis) and is identical work on the passthrough side (neutral for `G`).

3.5 Decoding

Decode reads the fields in the same order, rebuilds the matrix by `pivot_bits · coeffsᵀ` over GF(2) (a `uint8` matmul; the low bit of the wrapping sum is the parity), reattaches the prefix and leftover, truncates to `original_nbytes`, and checks the CRC and length. The encoder is deterministic (asserted byte-identical across re-runs for 18 frozen cases).

4. Preregistration

`docs/preregistration.md` is git-locked; the commit that adds it is an ancestor of every natural-corpus result. It fixes, before any natural-corpus number existed:

- **Meaningful positive, per file:** `G_pct ≥ 0.05` **and** `G ≥ 1024` bytes **and** exact round-trip **and** the container is a real deduction (`n_relations ≥ 1`, not passthrough) **and** the file is not a format-trick, derived-column, or checksum case. The 5 % floor is the midpoint of the 3–10 % band a compressor-generation change typically achieves; it is a relevance threshold, not a measurement-precision limit.
- **Hypothesis outcome:** *positive* iff ≥ 1 non-prior-art corpus meets the per-file threshold and reproduces within ±10 %; *negative* iff none does while the positive, null, and prior-art controls all behave as designed; *inconclusive* if a control fails or resource limits prevent running enough of the corpus list.
- **Kill criterion (three conjuncts):** negative on the full corpus list; **and** one bounded detector-broadening attempt moves no natural file above threshold; **and** the strongest executed context mixer does not itself close the planted-GF(2) gap. Its status is tracked in `docs/kill_criterion_status.md`.

Post-lock changes are limited to engineering that cannot move a reported size (codec vectorisation guarded by byte-exact equivalence tests, result-file schema additions, plotting). Baselines, presets, accounting rules, and thresholds are frozen.

5. Experimental protocol

5.1 Datasets and corpus coverage

The preregistered natural corpus is: the 12 Silesia members (whole files); enwik8 (first 10^8 bytes); ≥ 1 SDRBench scientific `float32` field; and the UCI Individual Household Electric Power Consumption text. The scientific and telemetry corpora were named in the lock, before any loader output was inspected, because they are the canonical public datasets for "exact cross-field structure is plausible here" — not because they produced a favourable number.

id	obtained	SHA-256 pinned	run as
`silesia_{dickens,xml,ooffice,reymont,sao,x-ray,mr,osdb}`	`silesia.zip` (Mahoney mirror)	yes	**whole** (5.3–10.2 MB)
`silesia_{samba,nci,webster,mozilla}`	same	yes	256 KiB prefix only
`enwik8`	`enwik8.zip`	yes	256 KiB prefix only
`sdrbench_exaalt2869440_{vx,vy,vz,xx,yy,zz}.f32`	SDRBench EXAALT bundle (Globus HTTPS), tar-verified	yes	256 KiB prefix (fields are 11.5 MB whole)
`uci_household_power_text`	UCI ML Repository #235	yes	256 KiB prefix only

Every corpus is pinned by SHA-256 in `results/corpus_manifest.json`; a changed hash aborts the run. No corpus has been dropped for being unfavourable; the four Silesia members not run whole are the *largest*, not the most inconvenient (§8). Prefixes are labelled as such and live in a separate result phase; they are never presented as whole-file results.

Train/test separation. The discovery step reads only the file it encodes; there is no learned model, no shared parameters across files, and no held-out tuning set. The only quantities fixed across all files are the six bit widths, the six stock compressors, and the preregistered threshold. Synthetic data is generated from documented integer seeds.

5.2 Baselines

`B = {gzip -9 (mtime 0), zlib -9, bz2 -9, xz -9 (FORMAT_XZ), zstd -19, brotli -11}`, all deterministic; versions recorded per record (`zstandard` 0.25.0, `brotli` 1.2.0, stdlib otherwise; Python 3.13.6). A compressor that raises `MemoryError` or times out is recorded unavailable with the exception and excluded from the `min`; the available set is reported per record.

Two context-mixing compressors — paq8l (`-3`, `-8`) and paq8px v216 (`-4`, `-8`) — are applied **only** to the planted-GF(2) positive control, to test whether the mechanism's advantage there is a genuine blind spot of the strongest statistical compressors we could run. cmix (needs 20–32 GiB, no memory level) and nncp (needs a GPU) could not be run on the available hardware; the paper writes "the strongest context-mixing compressors we could run", never "state of the art".

5.3 Independent verification

`verification/independent_verify.py` shares no code with the encoder or the main decoder: its own bit reader, its own container parse, a plain-Python XOR / integer reconstruction, and an independent re-derivation of the accounting from parsed field sizes. For every reported result it checks the full chain `raw → encode → container → compress → decompress → independent-decode → raw` and asserts SHA-256 equality and the accounting invariant. It reconstructs all four container kinds in a self-test, runs in CI over 21 parametrised cases, and was run on the 10 MB `silesia_dickens` container. The primary decoder's own composed round-trip is also checked for every stock compressor.

5.4 Reproducibility harness

`scripts/reproduce.py` runs, in order: `pytest` (630 tests: equivalence, randomised property tests, independent verifier), the verifier self-test, the controls, the natural-corpus slice sweep, the offset-extension sweep, the legacy phase experiments, then `build_ledger.py → regen_tables.py → make_figures.py → check_paper_numbers.py → independent_verify.py --ledger`. All ten steps return 0 on the reported state. Every experiment JSON records dataset SHA-256, seeds, config, the seven accounting categories, container size, round-trip and composed round-trip status, every baseline `(name, bytes, seconds, available)`, every composition size, `G`, `G_pct`, the raw gain, a UTC timestamp, the git commit (with a `-dirty` marker if the tree was not clean), and machine info.

6. Controls

All gate assertions pass (`experiments/controls/run.py`; full numbers in `paper/results_tables.md`).

control	construction	result
positive: planted GF(2), 3 seeds	random info + exact parity columns	`G_pct` +48.31 %, +49.5 %, +49.7 %; round-trip exact
null: i.i.d. bits / shuffled planted / 1-flip near-relation	—	passthrough; `|G| ≤ 19` bytes
corruption sweep, flip fraction `ϕ ∈ {0, 10⁻⁴, 10⁻³, 10⁻², 5·10⁻²}`	planted code, `ϕ·n_rows·n_parity` parity cells flipped	`G` decreases monotonically 16 213 → 13 238 → 1 402 → −18 → −18; never below −64
representation-change null	40 i.i.d. inputs through the full width × variant `min`	0 containers beat passthrough; 0 composed gains > 64 bytes
metadata-cost	null input; planted input	null overhead = exactly 18 bytes; planted container bits = Σ categories to the bit
composition-order	planted container through `decompress(compress(·))` for five codecs	primary decoder inverts every time
non-aligned period, `w = 48`	exact 48-bit-period linear code	default widths → passthrough (`G` -18); adding width 48 → 24 relations, round-trip exact
prior art: affine FD / per-record CRC32	derived column / CRC records	`G` = +37 880 / +16 209 bytes, **labelled** FD elimination / checksum inversion; not counted toward RQ-E

The corruption sweep and the non-aligned-period control together bound the scope of a negative: the method degrades gracefully as exact structure is broken, and it is blind to exact structure whose period is not among the tried widths (visible once the width is added).

7. Related work and novelty

The general idea — discover exact structure, transmit a reconstruction recipe, and beat a general-purpose compressor after counting the recipe — is not new.

- **Syndrome source coding** (Ancheta, 1976; and its universal generalisation). Treat the source as an error pattern and transmit its syndrome `Hx` under a linear code; a coset-leader source decompresses exactly. "In the absence of side information a Slepian–Wolf coder is an entropy coder." Compressing a single source by exploiting linear-code membership is classical; what is not classical is *discovering* the code from one input with description cost charged on the same channel — a thin distinction.

- **Grammar-based compression** (Kieffer–Yang; SEQUITUR; Re-Pair). The string is the unique yield of a grammar whose description cost is counted; structure is concatenative rather than linear-algebraic.
- **Functional-dependency / derived-column elimination.** US 8,150,888; Corra (2024); Wolpe (2026); TICC (2017); anisotropic columnar compression (US 11,562,085); Infobright-style inexact-FD-with-stored-exceptions. This space is thoroughly occupied for relational tables. Our affine-FD and CRC results are labelled as reproductions of it and are excluded from RQ-E.
- **Format-aware recompression.** Precomp, preflate, packJPG invert known codecs and drop reconstructible checksums; our PNG/ZIP/SQLite behaviour is the predicted format-awareness trap, not a counterexample.
- **Program-synthesis compression.** Brevis (2026) synthesises a self-contained DSL program that reconstructs a tensor bit-exactly and reports a real composed win on model checkpoints — a favourable domain dense with repeats, low rank and quantisation grids, searched with a rich language guided by a learned prior. This is a strong recent instance of the general idea; it does not occupy blind GF(2)/affine discovery on arbitrary bytes.
- **Low GF(2)-rank / Boolean matrix factorisation** (Fomin et al.; F_p-matrix factorisation). A change of basis to expose binary rank deficiency; NP-hard in general, and the exact case is the special case our column basis computes at a fixed width.
- **Invariant mining** (Daikon and successors). Templated discovery of equalities and affine relations among program variables; not a compressor and false-positive-prone.

Our contribution, stated narrowly. (i) An accounting-complete, never-worse GF(2)/affine relation codec with an independent shared-nothing verifier; (ii) the composed deduction-gain metric with a preregistered per-file threshold as an evaluation discipline for pre-pass compression claims; (iii) a preregistered empirical measurement — with a bounded phase-offset extension — of whether blind axis-aligned exact-relation discovery produces a composed advantage on public natural corpora, finding that it does not, on the corpora and detectors tested. We do not claim any of (i)–(iii) is "first" in a stronger sense than this.

8. Results

8.1 Mechanism validation (planted GF(2))

On random GF(2) linear codes — `k` information columns and `k` exact parity columns — the pre-pass behaves exactly as designed. The relation description is a constant of the code (1 088 bits for `k = 32`, 4,224 bits for `k = 64`), independent of the number of rows, so the composed gain grows with `n_rows`: `G_pct` rises from `+32.3 %` at 256 bytes to +49.95 % at 1 MiB (Fig. 1; Table B.2). Every case round-trips, and the independent verifier agrees.

Context mixers do not close this gap. On a 10 KiB code (seed 902), the best stock compressor leaves `+4 949` bytes recoverable by deduction; paq8l `-8` compresses the raw bytes to 10 321 and the container to 5 347 (a mixer-relative gap of `+4 974`), and paq8px v216 `-8` to 10 261 and 5 288 (`+4 973`). Raising the mixer memory level changes the gap by `< 6` bytes. The planted-GF(2) composed gain is therefore not an artifact of weak baselines; it is a blind spot of these statistical compressors on this bitstream. This is a statement about those compressors on that input, not a theorem about statistical models in general.

8.2 Natural corpora (RQ-C, RQ-D, RQ-E)

Whole files (8 of 12 Silesia members). No file meets the threshold (Table B.3). Three members admit GF(2) relations at a tried width but the container is *larger* than the best stock compressor's output, so RQ-C already fails; composition then makes it worse:

member	bytes	kind	rels	`raw_best`	`G` (bytes)	`G_pct`
`dickens`	10,192,446	GF2	25	2,799,520	-2,645,016	−94.5 %
`x-ray`	8,474,240	GF2	58	4,051,112	-1,907,747	−47.1 %
`xml`	5,345,280	GF2	21	430,390	-1,299,038	−301.8 %

The relations found on text/XML are the "high bit of a mostly-ASCII byte is zero" bit-plane and similar; brotli and bzip2 already exploit this from the raw byte order. The other five members admit no exact relation at any tried width and fall to passthrough, with `|G| ≤ 6 408` bytes (`|G_pct| ≤ 0.15 %`), i.e. downstream-compressor sensitivity to an 18-byte header, not deduction.

256 KiB files (20). Same pattern: 0 meaningful positives (Table B.4). Where GF(2) finds relations (`float32` fields, XML, text, telemetry) the composed gain is large-negative (`-4.8 %` to `-302 %`); where it does not, `|G_pct| < 0.3 %`. Across all 122 records the maximum `G_pct` is `+0.09 %` (`silesia_sao` at 256 KiB, a passthrough container), far below the 5 % floor and excluded anyway as passthrough.

Bit-phase-offset extension (20 files). For every file the offset-search composed gain equals the phase-0 value **to the byte** (Table B.5); no bit phase improves on phase 0 by more than header perturbation, and 0 files cross the threshold. The axis-aligned negative is robust to bit phase; it is not an artifact of reshaping from bit 0. This is the one bounded detector-broadening attempt the kill criterion requires, and it does not rescue the hypothesis.

8.3 Prior-art sanity cases

An integer affine derived column (`C = 3A + 5B + 7`, 8 192 rows) yields `G` = +37 880 bytes and a per-record IEEE CRC32 set yields `G` = +16 209 bytes (affine) / +15 701 bytes (homogeneous). These are labelled reproductions of functional-dependency elimination and checksum inversion respectively and are **not** counted toward RQ-E.

8.4 Verdict under the preregistered protocol

- Positive, null, prior-art, and scope controls all behave as designed.
- No non-prior-art natural file meets the per-file threshold, on 8 whole Silesia members, 20 files at 256 KiB, and the phase-offset extension.
- Four of the twelve Silesia members, and the whole enwik8 / SDRBench / UCI files, could not be run whole on the available 8 GiB machine (§9).

By the letter of the preregistration this is **inconclusive for the full corpus list** (the four large members are not yet run whole). For the coverage achieved it is a **clean, layered negative**: on natural data, exact structure is often present (RQ-A) and discoverable (RQ-B), and on no tested file does it reduce the representation (RQ-C) or survive composition (RQ-D), robustly to bit phase.

9. Discussion

9.1 What is established

- The codec recovers and exploits exact GF(2) linear structure when it is deliberately present: complete round-trip, complete accounting, a composed gain up to `+49.95 %` that scales with rows, independently verified end to end, and not closed by paq8l or paq8px v216.

- The controls establish that this behaviour is not a false positive of the width × variant search (0 spurious gains on 40 i.i.d. inputs), that metadata is fully counted (null overhead is exactly the 18-byte passthrough header), and that the pipeline degrades gracefully as structure is corrupted.
- On the natural corpora *actually measured* — 8 whole Silesia members plus 20 files at 256 KiB plus the phase-offset extension — the composed deduction gain is at most `+0.09 %` and is negative on every file where a real relation is found. The redundancy the detector locates in natural data is real but trivial (constant bit-planes, ASCII structure) and is already captured by stock compressors from the raw byte order.

9.2 What is not established

- **Not** that natural data contains no exact algebraic redundancy. We tested two specific relation families (axis-aligned fixed-width GF(2), integer affine on parsed tables), extended once to arbitrary bit phase, on one public benchmark corpus plus one scientific dataset (six correlated fields) plus one telemetry file. A materially different family — non-axis-aligned linear via a searched or learned column permutation, or low-degree polynomial relations — is untested.
- **Not** that the approach can never improve compression. Program-synthesis compression demonstrates a real composed win on a favourable domain (model checkpoints). Our negative concerns blind, prior-free, axis-aligned linear discovery on general byte corpora.
- **Not** that the method is universally inferior to existing compressors: on its target structure it wins decisively, and the never-worse guard makes it, by construction, at most 18 bytes worse than passthrough on any input.

9.3 Why the negative is informative

The mechanism is validated: when exact linear structure exists it is found and turned into a real, composition-surviving size reduction. The informative content is that the *analogous* structure a blind axis-aligned detector locates in natural byte corpora is (i) already modelled by stock compressors and (ii) too cheap-to-describe-relative-to-what-it-recovers to survive its own metadata, let alone composition. Establishing a positive on natural data would require either a corpus with genuine exact cross-field structure that stock compressors miss (the FD/CRC cases show what that looks like, and they are prior art), or a detector family expressive enough to find non-trivial structure at acceptable description cost — the direction future work would have to take, and the one the preregistered kill criterion names.

10. Threats to validity

threat	mitigation
encoder and decoder share a bug	independent shared-nothing decoder + independent accounting re-derivation, in CI and on the 10 MB `dickens` container
vectorised primitives are wrong	400 randomised bit-I/O trials vs a per-bit reference; 60 `reconstruct` trials vs a per-column XOR reference; 18 byte-frozen container cases
a container has unaccounted bits	finaliser refuses to emit it; independently re-derived by the second decoder
the downstream comparison is unfair (a codec fails on one side)	matched-codec gain over the intersection; both available-codec sets recorded; a reportable positive needs matched sets
the compressed container does not decode back	`raw → encode → compress → decompress → decode → raw` checked with both the primary and the independent decoder for every codec; 0 failures over 122 records
the width × variant `min` inflates results	representation-change null: 0 spurious gains on 40 i.i.d. inputs; never-worse guard caps the downside at passthrough
the negative is a framing / bit-phase artifact	bounded phase-offset extension on all 20 files: `G` equals the phase-0 value to the byte
a prefix is passed off as a whole file	separate result phase, separate manifest keys, `prefix` reason on every row
the corpus was chosen after seeing results	corpus list git-locked before any loader output; every file SHA-256-pinned; a changed hash aborts
numbers were transcribed by hand	all tables generated from `results/ledger.json`; inline figures carry markers checked by `check_paper_numbers.py`

11. Limitations

- **Detector scope.** Axis-aligned fixed-width GF(2) plus integer affine, extended once to every bit phase. A negative is scoped to these families; the non-aligned-period control shows exact structure at a period not among the tried widths is missed by construction.
- **Whole-file coverage.** 8 of 12 Silesia members run whole (≤ 10.2 MB); the four largest (21–51 MB) and whole enwik8 / SDRBench / UCI need more than 8 GiB RAM than the development machine has. Every 256 KiB slice, every whole file run, and the offset extension agree, so the *conclusion* is consistent, but the pre-registered protocol is *not complete* for the full list.
- **Baseline ceiling.** cmix and nncp were not run (hardware). The planted-GF(2) result with paq8l/paq8px bounds — but does not eliminate — the possibility that a stronger mixer would absorb the planted gap; it says nothing about the natural-corpus negative, which does not depend on baseline strength.
- **Corpus breadth.** Four corpus classes; the scientific class is one molecular-dynamics dataset whose six fields are one simulation (effective n ≈ 1–2). A broader scientific and telemetry sweep is future work.
- **Statistics.** The measurements are deterministic; there is no random sampling from a defined population, so no inferential confidence interval or p-value is claimed. The file is the unit of analysis and the headline is a per-file maximum `G_pct` against the fixed threshold.

12. Conclusion

We asked whether automatically discovered exact algebraic relations can provide an additional source of lossless compression that survives full description cost and composition with a strong downstream compressor. We built an accounting-complete GF(2)/affine relation codec, an independent verifier, and a composed-gain metric, and we preregistered the evaluation. The mechanism is validated on deliberately structured data: it recovers planted GF(2) codes and converts them into a composed size reduction of up to `+49.95 %` that scales with data size and is not closed by two context-mixing compressors. Under the preregistered protocol, the analogous exploitable structure does **not** appear on the natural corpora tested — 8 whole Silesia members, 20 further files at 256 KiB, and a bounded bit-phase-offset extension all yield no composed gain meeting the threshold, with a maximum observed `G_pct` of `+0.0009`. The functional-dependency and CRC cases reproduce known techniques. The mechanism is not novel; the contribution is the accounting discipline, the composed evaluation, the controlled validation, and the preregistered empirical finding. The result is a scoped negative, not a refutation of the idea: it establishes that a blind, prior-free, axis-aligned linear detector does not expose a composed advantage on general byte corpora, and it makes explicit what a future positive would have to establish.

13. Reproducibility

Environment: Windows 11 (10.0.22631), 4 logical cores, 8 GiB RAM; Python 3.13.6; numpy 2.2.6; zstandard 0.25.0; brotli 1.2.0. Corpora are not committed (licence and size); acquisition is scripted and SHA-256-pinned, and a changed hash aborts.

```
python -m pip install -e ".[dev]"
python -m pytest -q                       # 630 tests
python verification/independent_verify.py --self-test

python experiments/controls/run.py        # exit 0 == all gates pass
python experiments/natural/run.py  --mode whole        # >= 32 GiB machine: the full answer
python experiments/natural/run.py  --mode whole --only silesia_xml   # one file at a time (8 GiB ok to ~10 MB)
python experiments/offset/run.py   --mode slice        # bit-phase-offset extension

python scripts/build_ledger.py && python scripts/regen_tables.py && python scripts/make_figures.py
python scripts/check_paper_numbers.py
python verification/independent_verify.py --ledger results/ledger.json

python scripts/reproduce.py --mode whole --offset-whole   # everything, big machine
```

On the reported state every step returns 0; a re-run reproduces every size, gain, and verdict exactly, with only timings and timestamps differing. `docs/preregistration.md` (locked), `docs/audit.md` (implementation audit and corrections C1–C2), `docs/prior_art.md`, `docs/statistics.md`, `docs/kill_criterion_status.md`, and `docs/venue_assessment.md` accompany the manuscript.

Appendix A — Figures

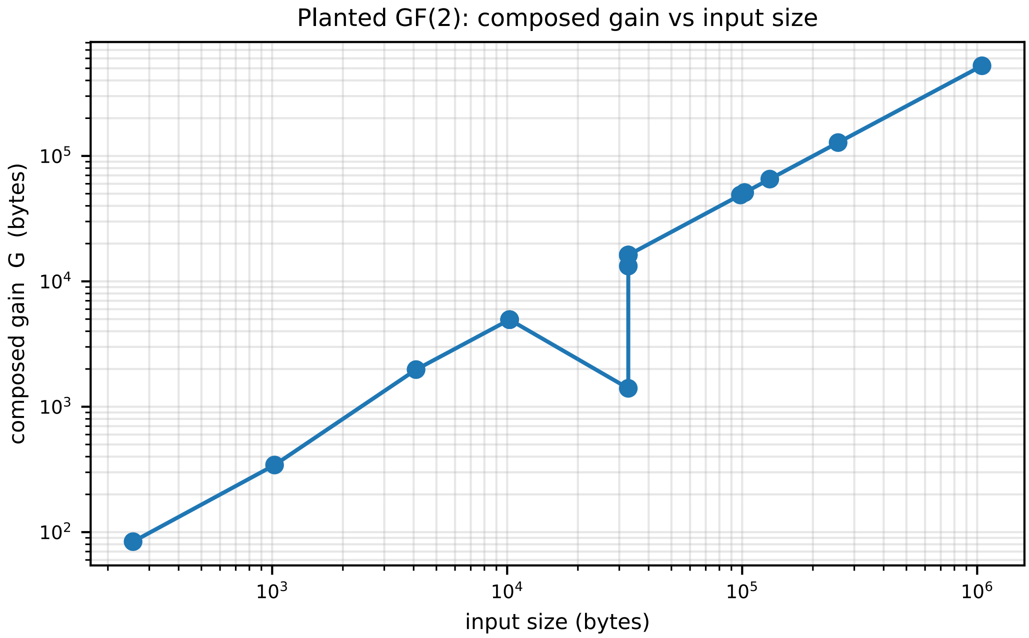

Figure 1. Planted GF(2) codes: composed gain `G` (bytes) versus input size, log–log. The gain scales approximately with the number of rows because the relation description is a constant of the code. The two low points are a single-flip near-relation control and an affine-parity control.

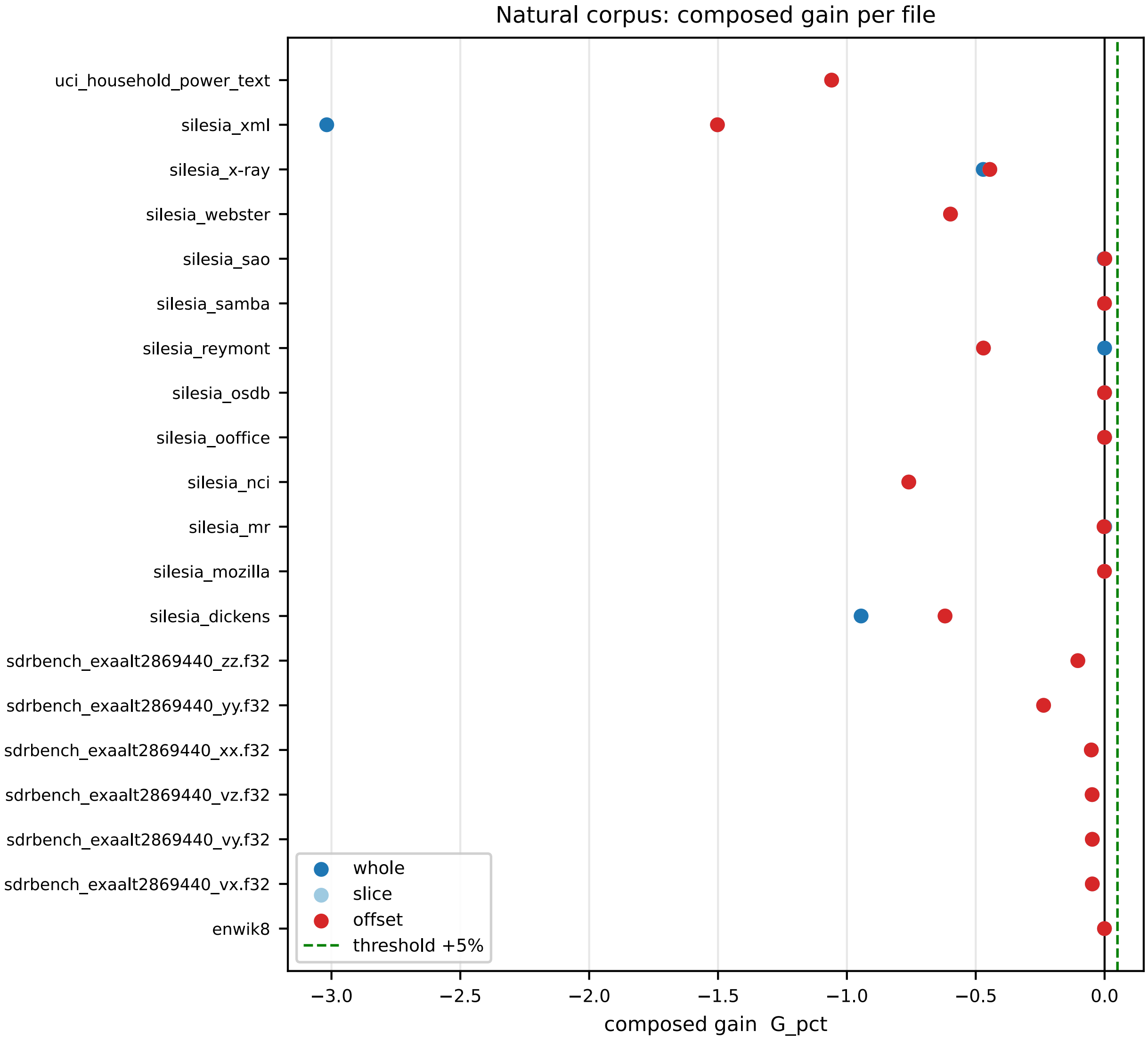

Figure 2. Natural corpus: per-file `G_pct` for the whole files, the 256 KiB slices, and the bit-phase-offset extension, with the zero line and the preregistered `+5 %` threshold. Every point is at or below zero.

Appendix B — Key results tables

The complete tables (**B.1** controls, **B.2** planted scaling + context-mixer baseline, **B.3** whole natural, **B.4** 256 KiB natural, **B.5** bit-phase offset, **B.6** prior art) are auto-generated in `paper/results_tables.md` from `results/ledger.json` (122 records; 0 accounting, round-trip, or composed round-trip failures). Every row traces to a JSON record under `results/` carrying its dataset hash, git commit, configuration, compressor versions, and verification status. The three most load-bearing are reproduced here.

B.2 (extract) — planted GF(2): the mechanism scales

input bytes	relations	container	`raw_best`	`G` (bytes)	`G_pct`	relation bits
256	8	172	260	+84	+32.3 %	80
4,096	16	2,118	4,100	+1,978	+48.2 %	288
32,768	32	16,554	32,772	+16,214	+49.5 %	1,088
102,400	32	51,371	102,405	+51,030	+49.8 %	1,088
256,000	32	128,171	256,005	+127,829	+49.9 %	1,088
1,048,576	64	524,850	1,048,581	+523,726	+49.95 %	4,224

B.3 — whole natural files (8 of 12 Silesia members)

member	bytes	kind	rels	`raw_best`	container	`G` (bytes)	`G_pct`	round-trip
`xml`	5,345,280	GF2	21	430,390	4,907,484	−1,299,038	−301.8 %	✓
`ooffice`	6,152,192	passthrough	0	2,426,816	6,152,210	−188	−0.0 %	✓
`reymont`	6,627,202	passthrough	0	1,246,230	6,627,220	+217	+0.0 %	✓
`sao`	7,251,944	passthrough	0	4,415,072	7,251,962	−6,408	−0.15 %	✓
`x-ray`	8,474,240	GF2	58	4,051,112	6,555,798	−1,907,747	−47.1 %	✓
`mr`	9,970,564	passthrough	0	2,441,280	9,970,582	+857	+0.0 %	✓
`osdb`	10,085,684	passthrough	0	2,802,792	10,085,702	−25	−0.0 %	✓
`dickens`	10,192,446	GF2	25	2,799,520	9,197,882	−2,645,016	−94.5 %	✓

B.5 (summary) — bit-phase-offset extension (20 files, 256 KiB). For every file the offset-search `G` equals the phase-0 `G` to the byte; 0 files cross the `+5 %` threshold; 0 files improve on phase 0 by more than header perturbation. The axis-aligned negative is robust to bit phase.